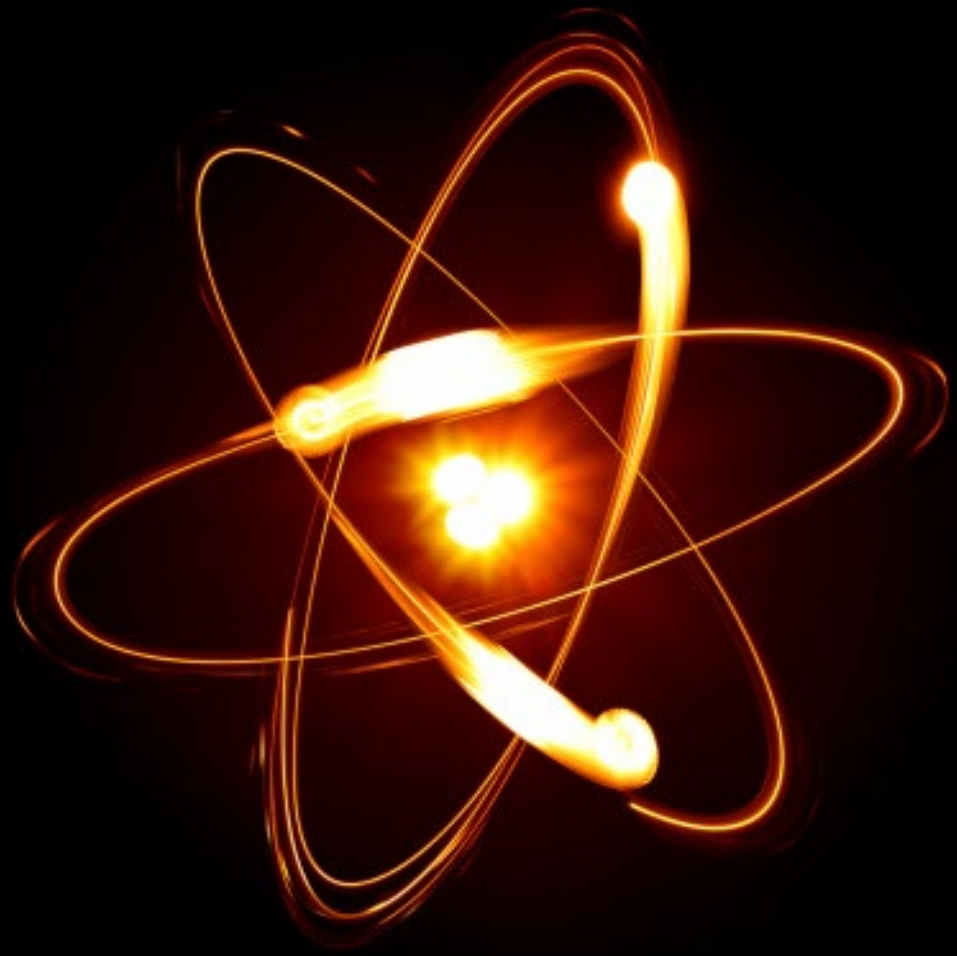
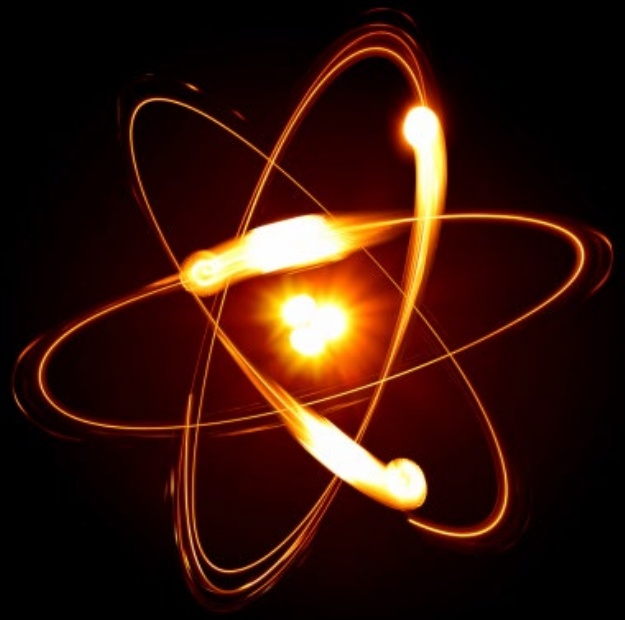
Chapter 18.

Orbital angular momentum



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Quantum Physics



Chapter 18.

Orbital angular momentum

- Orbital angular momentum. Introduction
- Orbital angular momentum. Commutation rules
- Orbital angular momentum. Eigenfunctions and eigenvalues
- Orbital magnetic dipole moment
- Stern-Gerlach experiment

Orbital angular momentum

We study now the angular momentum, applying the correspondence principle. This magnitude has great importance both in classical and in quantum physics.
Classically we have

$$\vec{l} = \vec{r} \wedge \vec{p}$$

According to the correspondence principle, the quantum angular momentum operator will be:

$$\vec{L} = \vec{R} \wedge \vec{P}$$

This is a vector operator, whose cartesian components are:

$$\vec{l} = \vec{r} \Lambda \vec{p} = \vec{i}\left(yp_z - zp_y\right) + \vec{j}(zp_x - xp_z) + \vec{k}\left(xp_y - yp_z\right)$$

Orbital angular momentum

Classical	Quantum
$l_x = yp_z - zp_y$	$L_x = YP_z - ZP_y$
$l_y = zp_x - xp_z$	$L_y = ZP_x - XP_z$
$l_z = xp_y - yp_x$	$L_z = XP_y - YP_x$
$l^2 = l_x^2 + l_y^2 + l_z^2$	$L^2 = L_x^2 + L_y^2 + L_z^2$

Before solving for the eigenvalues of these operators we examine the compatibility between measurements of the corresponding physical magnitudes.

Orbital angular momentum. Commutation rules

It is very easy to check that:

$$[L_x, L_y] = i\hbar L_z$$
$$[L_y, L_z] = i\hbar L_x$$
$$[L_z, L_x] = i\hbar L_y$$
$$[L^2, L_x] = [L^2, L_y] = [L^2, L_z] = 0$$



we can measure simultaneously:

L^2 and L_x

L^2 and L_y

L^2 and L_z

Orbital angular momentum. Commutation rules

$$\boxed{[L_x, L_y]} = L_x L_y - L_y L_x =$$

$$(YP_z - ZP_y)(ZP_x - XP_z) - (ZP_x - XP_z)(YP_z - ZP_y) =$$

$$= YP_zZP_x - YP_zXP_z - ZP_yZP_x + ZP_yXP_z - ZP_xYP_z + ZP_xZP_y + XP_zYP_z - XP_zZP_y =$$

$$= P_zZ(YP_x - XP_y) - ZP_z(P_xY - P_yX) = (P_zZ - ZP_z)(YP_x - XP_y) = \boxed{i\hbar L_z}$$

The other commutators are obtained in a similar way.

Orbital angular momentum. Eigenfunctions and eigenvalues

Now we solve for the eigenvalues and eigenfunctions of both L^2 and L_z.

We first find the expression of the angular momentum operators in position representation and change to spherical coordinates, taking into account the definition of the linear momentum and the change of variables from cartesian to spherical.

It is readily found that:

$$L_x = i\hbar\left(\sin\phi\frac{\partial}{\partial\theta} + \cot\theta\cos\phi\frac{\partial}{\partial\phi}\right)$$

$$L_y = i\hbar\left(-\cos\phi\frac{\partial}{\partial\theta} + \cot\theta\sin\phi\frac{\partial}{\partial\phi}\right)$$

$$L_z = -i\hbar\frac{\partial}{\partial\phi}$$

$$L^2 = -\hbar^2\left[\frac{1}{\sin\theta}\frac{\partial}{\partial\theta}\left(\sin\theta\frac{\partial}{\partial\theta}\right) + \frac{1}{\sin^2\theta}\frac{\partial^2}{\partial\phi^2}\right]$$

Orbital angular momentum. Eigenfunctions and eigenvalues

When we studied the Schrödinger equation for a particle in a spherically symmetric potential we separated the problem into a radial and an angular equation. The latter read:

$$-\frac{\hbar^2}{2\mu}\left[\frac{1}{\sin\theta}\frac{\partial}{\partial\theta}\left(\sin\theta\frac{\partial}{\partial\theta}\right)+\frac{1}{\sin^2\theta}\frac{\partial^2}{\partial\phi^2}\right]Y=\lambda Y$$

$$-\hbar^2\left[\frac{1}{\sin\theta}\frac{\partial}{\partial\theta}\left(\sin\theta\frac{\partial}{\partial\theta}\right)+\frac{1}{\sin^2\theta}\frac{\partial^2}{\partial\phi^2}\right]Y=2\mu\lambda Y$$

The angular equation can therefore also be written as:

$$L^2Y=2\mu\lambda Y$$

We already know the solutions of the angular equation.

Orbital angular momentum. Eigenfunctions and eigenvalues

Namely, the spherical harmonics, which read:

$$Y_l^m(\theta,\phi) = \left[\frac{(2l+1)(l-|m|)!}{4\pi(l+|m|)!}\right]^{\frac{1}{2}} P_l^{|m|}(\cos\theta)e^{im\phi}$$

with $|m| \leq l$ and $\lambda = \frac{\hbar^2}{2\mu} l(l+1)$, with $l = 0,1,2,3, \ldots$

$$L^2 Y_l^m = \hbar^2 l(l+1) Y_l^m$$

Therefore the eigenfunctions of L^2 are the spherical harmonics and the corresponding eigenvalues are $\hbar^2 l(l+1)$ with $l = 0,1,2,3 \ldots$

Are the spherical harmonics also eigenfunctions of L_z ?

$$L_z Y_l^m = -i\hbar \frac{\partial}{\partial\phi} Y_l^m = m\hbar Y_l^m$$

The spherical harmonics are also eigenfunctions of L_z with eigenvalues $m\hbar$.

Orbital angular momentum

Conclusions

- The square of the angular momentum is quantized, and can only take values that are some integers (of the form $l(l+1)$) times $\hbar^2$.

- The *z* component of the angular momentum is also quantized, and can only take values that are an integer times $\hbar$.

- It is physically impossible to know the angular momentum (the vector) of a particle, since the three components do not commute with each other, and therefore they cannot be measured simultaneously.

- The most we can know (measure simultaneously) is the module (or its square) and one component of the angular momentum vector.

Orbital angular momentum

Spherically symmetric potential and angular momentum.

The hamiltonian for a particle in a spherically symmetric potential was:

$$H = -\frac{\hbar^2}{2\mu}\left[\frac{1}{r^2}\frac{\partial}{\partial r}\left(r^2\frac{\partial}{\partial r}\right)\right] + V(r) - \frac{\hbar^2}{2\mu}\left[\frac{1}{r^2\sin\theta}\frac{\partial}{\partial\theta}\left(\sin\theta\frac{\partial}{\partial\theta}\right) + \frac{1}{r^2 sin^2\theta}\frac{\partial^2}{\partial\phi^2}\right]$$

which can be rewritten as,

$$H = -\frac{\hbar^2}{2\mu}\left[\frac{1}{r^2}\frac{\partial}{\partial r}\left(r^2\frac{\partial}{\partial r}\right)\right] + V(r) + \frac{L^2}{2\mu r^2}$$

We immediately see that $[H, L^2] = 0$, and $[H, L_z] = 0$, and it is therefore possible to find eigenfunctions common to H, L^2 and L_z, of the form $R(r)Y_l^m(\theta,\phi)$

Orbital angular momentum and magnetic properties[1]

Experimental measurement of the orbital angular momentum

Consider a one-electron atom. We want to measure the orbital angular momentum $\vec{L}$ of the electron.

This is achieved by measuring its orbital magnetic dipole moment $\vec{\mu}$ and analyzing its interaction with an external magnetic field.

Relation between $\vec{\mu}$ and $\vec{L}$?

Charged particle (q) with mass m and moving with velocity v in a circular Bohr orbit of radius r

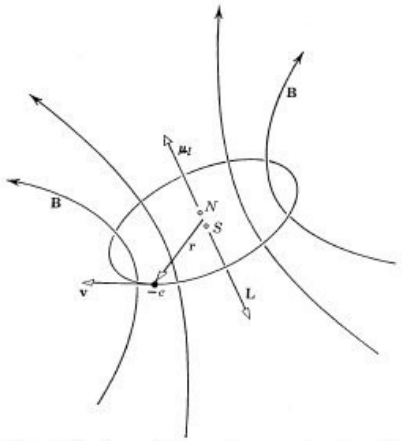


Figure 8-1 The orbital angular momentum **L** and the orbital magnetic dipole moment μ_l of an electron $-e$ moving in a Bohr orbit. The magnetic field **B** produced by the circulating charge is indicated by the curved lines. The fictitious magnetic dipole that would produce an identical field far from the loop is indicated by its poles *N*, *S*.

This amounts to a current of magnitude

$$i = q/T = qv/2\pi r$$

[1]Eisberg-Resnick page 267

Orbital magnetic dipole moments

This current produces a magnetic field, equivalent to that of a magnetic dipole located at the center of the loop and perpendicular to the plane.

The orbital magnetic dipole moment of the dipole is $\vec{\mu} = \frac{q}{2m}\vec{L}$ where $|\vec{L}| = mvr$.

For an electron $q = -e$ and it is usually written as

$$\vec{\mu} = \frac{-e}{2m_e}\vec{L} = -\frac{g_l \mu_b}{\hbar}\vec{L}$$

where $g_l(= 1)$ is the orbital g-factor and $\mu_b = \frac{|e|\hbar}{2m_e}$ is Bohr magneton.

$\vec{\mu}$ is a measurement of the strength of the magnetic dipole.

μ_b is usually used as unit for the measurement of atomic magnetic dipole moments.

Orbital magnetic dipole moments

If this magnetic dipole is placed inside a magnetic field $\vec{B}$, then it is subjected to a force with an associated potential energy

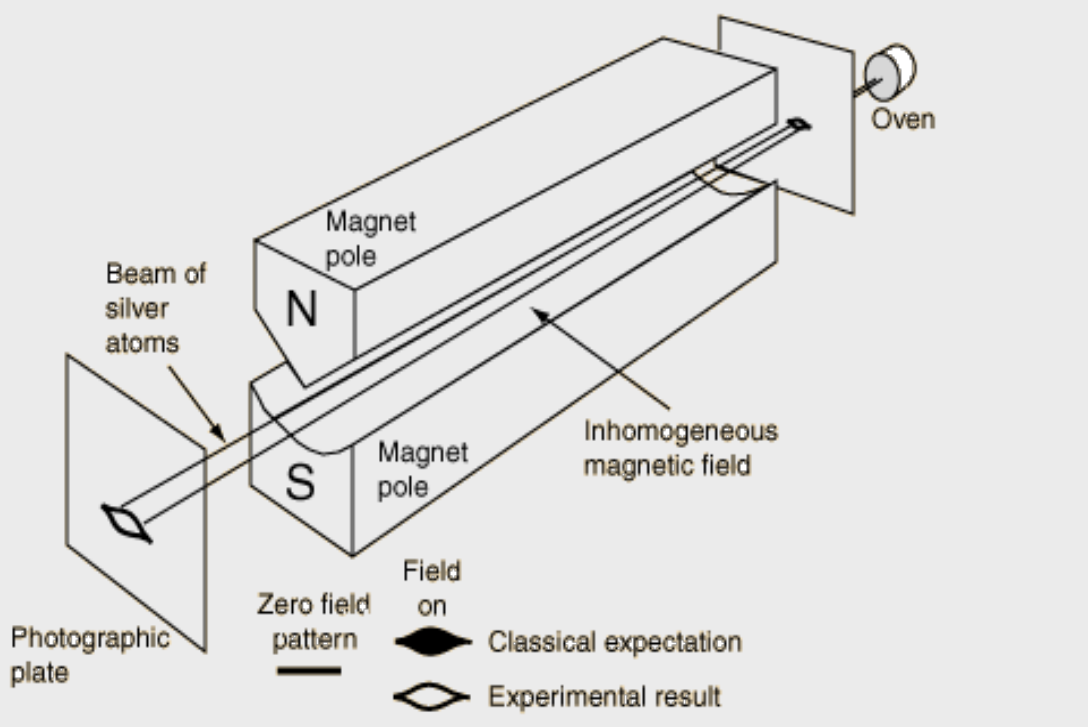
$$\vec{F} = \nabla(\vec{\mu}\vec{B}) \qquad \Delta E = -\vec{\mu}\vec{B} = \frac{g_l \mu_b}{\hbar}\vec{L}\vec{B}$$

If $\vec{B} = (0,0,B_z)$, then $\Delta E = \frac{g_l \mu_b}{\hbar} L_z B_z$ and the hamiltonian of the system (electron under an external potential) will have an extra term.

The extra term includes L_z. Its values are quantized so the same applies for the associated energy values. If we measure the effect of this extra term we may obtain information about the values of L_z.

Are the experimental results in agreement with the theoretical ones?

Stern-Gerlach experiment



1922. Measurement of the magnetic dipole moment for (neutral) silver atoms.

A beam of atoms is sent along the y-axis. There is a magnetic field $\vec{B} = (0,0,B_z)$, which changes along the z-axis.

Each atom has a magnetic dipole moment (due to the electrons) such that $\vec{\mu} = \gamma \vec{L}$. When the atoms are inside the magnetic field, there is a force acting on them

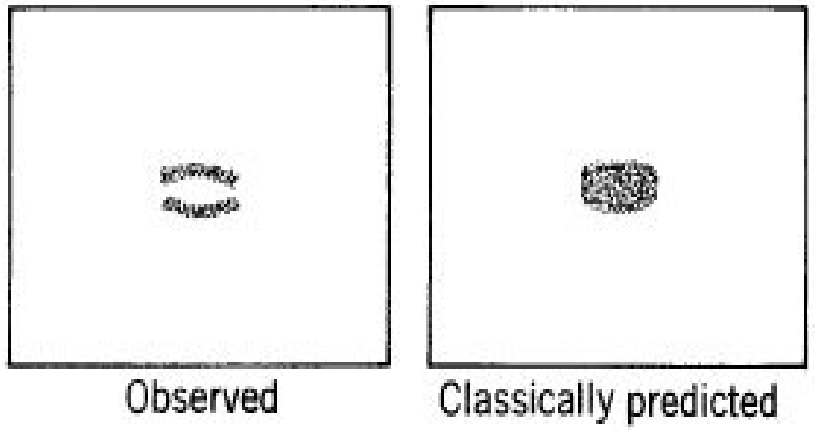
$$\vec{F} = \nabla(\vec{\mu}\vec{B}) = \mu_z \frac{\partial B_z}{\partial z}\vec{k}$$

The force is proportional to L_z.

CLASSICAL. Any value is allowed (within the range -L, L). The atoms are deflected leaving a continuous trace.

QUANTUM. We know that there are a discrete number of values for L_z. So we will observe some spots (2l+1).

Stern-Gerlach experiment



The beam of silver atoms splits into TWO components. It means two possible values for μ_z OR for L_z.

But for an orbital angular moment, it always has an odd number of allowed L_z values. !!! SOMETHING IS MISSING!!!!

Experiment with hydrogen atoms in ground state. Only one electron and l=0, m_l=0 and therefore μ_l = 0.

Implies NO splitting should appear.

BUT... we observe TWO components.

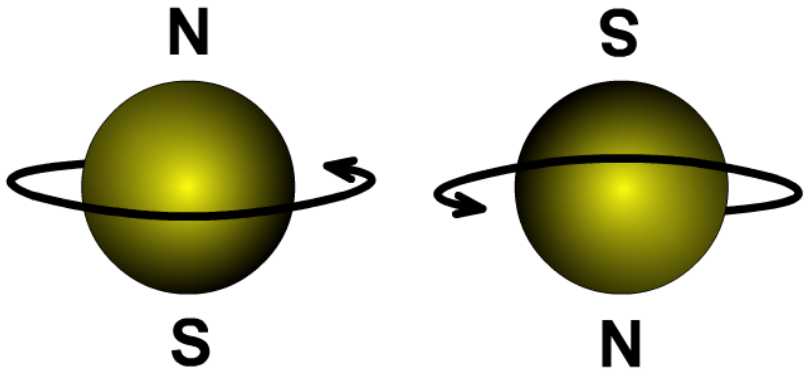
Stern-Gerlach experiment

EXPLANATION.

Assume the electron has an intrinsic magnetic dipole moment, which is induced by an intrinsic angular momentum $\vec{S}$ (spin).

Assume the same relation

$$\vec{\mu}_s = -\frac{g_s \mu_b}{\hbar}\vec{S}$$



which means $\mu_{sz} = -\frac{g_s \mu_b}{\hbar} m_s$

Experiment gives TWO components.

Then m_s must have two values, which implies s=1/2 and therefore $m_s = \pm 1/2$.

Here g_s = 2 and it is called the spin g-factor